

Qilmeg Doudatz

Cambridge, MA · qimg954@mit.edu · (734) 546-7287 · qilmegd.com

Researcher and designer specializing in 3D neural scene representations, leveraging human spatial behavior in digitized environments to advance human-centric computer vision.

EDUCATION

Massachusetts Institute of Technology	Expected 06/27
M.S. Electrical Engineering & Computer Science (SM EECS)	GPA 4.8 / 5.0
Dual degree with M.S. in Architecture Studies (SMArchS), Computation Group	
Advisor: Prof. Takehiko Nagakura · Committee: Prof. William T. Freeman	
University of Michigan, Taubman College	Master of Architecture (M.Arch) 05/24
Southeast University, Nanjing	B.Eng. in Civil Engineering 06/21
Coursework	MIT 6.S058 Computer Vision · 6.8370 Advanced Computational Photography · 6.7960 Deep Learning · 6.C511 Modeling with Machine Learning · 9.660 Computational Cognitive Science · 6.9020 How to Make Almost Anything (also TA)

PUBLICATIONS AND PREPRINTS

Doudatz, Q., & Nagakura, T. (2026). Designing AI to design experiments for studying humans in spatial design: framework and case study. ACADIA 2026 (accepted).

Gallardo, R., **Doudatz, Q.**, Goldstein, G., Sarimehmetoglu, I., Mutis, S., Kyaw, A. H., Lin, A., Emmerling, C., Ataman, B., & Tibbits, S. (2026). Context-aware AI assistant and AR interface for lunar extravehicular activity procedural guidance. SIGGRAPH '26 Posters. ACM.

Ataman, B., Gallardo, R., & **Doudatz, Q.** (2026). Affective translation: emotional engagement with textile soft robots across physical and augmented embodiments. HCI International 2026 Posters, 3–12. Springer. **Best Poster Award.**

Doudatz, Q., & Nagakura, T. (2026). Morphic Memory: a crowdsourced vision-language framework for documenting vernacular architectural intelligence. SiGraDi 2026 (accepted).

Ataman, B., Gallardo, R., & **Doudatz, Q.** (2025). Affective translation: material and virtual embodiments of kinetic textile robots. arXiv:2601.11543 [cs.HC].

Doudatz, Q. (2025). Vertönts, the vertical world. Dimensions, 38(1), 80.

Tuladhar, S.*, **Doudatz, Q.***, & Samuelson, H. (in preparation). Measured thermal comfort performance of custom rice-husk insulation in cold-climate emergency shelters. Target: Building and Environment. (* equal contribution)

RESEARCH EXPERIENCE

MIT Architecture Representation & Computation Group	Graduate Research Assistant	Jan 2026 – Present
- Constructed aligned multi-representation captures of the same space in VR (Gaussian splats, mesh, point cloud) with semantic segmentation over each. - Training models for intent inference and representation selection that switch what the headset renders in real time, from gaze and movement telemetry captured in a VR attention study I designed and ran with 19 participants. - Improved novel view synthesis from 360° capture by evaluating three camera-model treatments of dual-fisheye footage: raw fisheye through 3DGUT's unscented transform, pinhole reprojection, and tiled equirectangular. - Built the dataset pipeline for splat training, using SAM 2 to segment and remove people from fisheye and 360° video before structure-from-motion.		
Livable Space Lab, MIT	Graduate Research Assistant	Aug 2025 – Present
- Building exhibition web application (Venice ECC) letting audience tablets drive a synchronized gallery display. - Instrumented it as a data collection pipeline, turning visitor interactions into labeled human preference data for fine-tuning a window-design automation model. - Co-first author on a measured study of rice-husk insulation performance in emergency shelters (in preparation).		
Simulated Assemblies Lab, UMich	Research Assistant & Game Designer	Sep 2024 – Jun 2025
Built the player behavior analysis pipeline for a research game, logging positional and timestep data and analyzing movement patterns to inform mechanic design.		

- Empathy in Point Clouds Scan Lab, UMich**

Visualization Specialist — FARO LiDAR capture

Aug 2023 – Jun 2024
- Digitized cultural heritage sites across Inner Mongolia, Mexico, Michigan, and Arizona with terrestrial LiDAR and photogrammetry, and built interactive game-engine walkthroughs of the scanned data.
- DESIGN EARTH, UMich**

Research Assistant

Jan 2023 – Jun 2024
- Developed the practice's portfolio site and the companion website for A World Previous to Ours at the Venice Architecture Biennale, extending the exhibition with contextual material for visitors.
- More Room at the Table, UMich**

Research Assistant

Aug 2022 – May 2023
- Built gesture-interactive VR environments in Unity that manipulate perceived depth, applying forced-perspective constructions in the tradition of the Ames room and Teatro Olimpico.

INDUSTRY EXPERIENCE

- Glydways**

Software Engineer Intern, Perception

Jun – Sep 2026
- Shipped the first camera-based simulation environments for a LiDAR-only perception stack by reconstructing 3D Gaussian Splatting scenes from raw vehicle logs across six track segments.
 - Built the preprocessing stage controlling frame sampling and lens distortion before training, improving reconstruction quality across all six logs (PSNR above 23 dB on held-out views).
 - Selected the production pipeline by benchmarking ~6 open-source frameworks down to two finalists (OmniRe/DriveStudio, SplatAD/NeuRAD) on camera and LiDAR rendering and object editing.
 - Collapsed a three-day manual process into one command running unattended overnight, and cut GPU cost per training run ~5x by migrating from AWS A10G to A100s on Thunder Compute.
- University of Michigan Center for Academic Innovation**

3D Artist Fellow

Jan 2024 – May 2024
- Produced 3D environments in Unreal Engine for LED volume virtual production, compositing digital sets with live action for educational film.

AWARDS AND HONORS

HCI International 2026, Best Poster Award	2026
NASA Spacesuit User Interface Technologies for Students (SUITS) Challenge, Winner (Team PR Lead)	2026
MIT Web Lab Competition, Most Unique Concept Award	2026
MIT School of Architecture and Planning Department Fellowship	2025
Burton L. Kampner Memorial Award (Taubman College Best Thesis)	2024
RIBA President's Medals, Silver Medal Category, Nominee	2024
University of Michigan Arts Initiative Graduate Student Arts Funding Grant	2023
Arts Integrative Interdisciplinary Research (AiIR) Grant, Arts Engine	2023
University of Michigan XR Student Showcase, Best Use of XR for Research	2023
Southeast University Distinguished Bachelor's Thesis Award	2021

TEACHING AND SERVICE

Teaching Assistant, MIT 6.9020 How to Make Almost Anything	2026
Teaching Assistant, ARCH 314/324 Structures I–II, UMich	2023 – 2024
Workshop Instructor, ARCH 509 Point Clouds and Unreal Worlds; ARCH 662 Blender Workshop, UMich	2023 – 2024
Invited Reviewer, UMich design studios (ARCH 432, 442, 662, 672)	2023 – 2024

SKILLS

Languages & Systems	Python, C++, Bash, JavaScript/Node, HTML/CSS, Linux (Ubuntu), WSL2, Git
Computer Vision & 3D	Structure-from-motion (COLMAP), 3D Gaussian Splatting, camera models and distortion (pinhole, fisheye, equirectangular), camera–LiDAR registration, terrestrial LiDAR, SAM 2
ML & Image Processing	PyTorch, CNNs, vision transformers (ViT, VideoMAE), model fine-tuning, HDR processing, multimodal LLM and vision-language models, OpenCV, NumPy, FFmpeg
Real-Time & Web	Unity, Unreal Engine 5, OpenXR, Three.js, WebXR, React, Node